Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/02/2018
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Changed JPL Task Plan number to 71-15884 per U. Israelsson/JPL (Ed., 5/27/2020) NOTE: End date changed to 9/30/2022, per PI (Ed., 2/27/2020) NOTE: End date changed to 9/30/2019, per PI (Ed., 11/17/17); changed to 10/1/2019 to accommodate final reporting (Ed., 10/4/18) NOTE: End date changed to 10/01/2017 (Ed., 11/14/16) NOTE: Extended for reflight to 9/30/2017 (per PI and U. Israelsson/JPL)-Ed., 5/12/14 NOTE: Extended to 3/31/2014 per PI (Ed., 10/21/13)		
Key Personnel Changes/Previous PI:			

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Grant/Contract No.:	JPL Task Plan number 71-15884 ; JPL Task Plan number 71-15890
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologists, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning Renormalization Group (RG) theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on "seemingly innocuous" assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient. NASA's MISTE/COEX (microgravity scaling theory experiment/coexistence curve experiment) flight experiments were high precision studies of static critical phenomena planned on board the International Space Station (ISS). Those experiments were to address one of the most difficult experimental challenges in studying static properties of near critical fluid in microgravity, which is associated with the equilibration time of the sample. Recently, the DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques) facility supported by CNES/NASA was successfully integrated into the ISS. The DECLIC facility developed by CNES (Centre national d'études spatiales) investigators allows precision studies of non-equilibrium phenomena near the liquid-gas critical point. Multiple science inserts for the DECLIC facility have been developed to study room temperature and high temperature critical fluids. These inserts are currently on-board ISS to study the boiling phenomena, phase separation, and piston effect of near critical fluids. The DECLIC/ALI experiment is to be performed in the near future. Detailed discussions with the DECLIC Principal Investigators (PIs) have clearly indicated that additional important scientific investigations can be carried out utilizing the available DECLIC/ALI hardware. Taking advantage of the current CNES funded DECLIC facility, we propose to NASA to study the temperature and density relaxation behavior of the fluid (SF6) near its liquid-gas critical point by utilizing DECLIC/ALI experiments. By analyzing experimental data taken under microgravity conditions, critical data to understand the equilibration processes of a near critical fluid, especially in the two-phase region, can be obtained that is not feasible on the Earth. The results of data analysis, modeling, simulations, and ground support experiments will enhance the scientific return of the proposed DECLIC investigations. This study of equilibration near a critical fluid will directly support the objectives of NASA's MISTE/COEX flight experiment and benefit future critical phenomena studies utilizing the DECLIC facility. The research team of this proposal consists of NASA MISTE/COEX investigators and CNES DECLIC principal investigators. APRIL 2014--ADDITIONAL INFORMATION FOR REFLIGHT PROJECT (ALI-R) A re-flight of the DECLIC Alice-Like-Insert (ALI-R) under microgravity conditions is to measure the light transmission and the turbidity of sulfurhexafluoride (SF6) fluid in the single-phase region. The new experiment will utilize the existing DECLIC facility, the ground control facility at CNES, and a new measurement cell. The ALI-R will be able to experimentally measure the effects of the non-zero value of the Green-Fisher exponent on the liquid-gas critical point for the first time. A measurement of the turbidity of the SF6 sample with a precision of 0.3% at ~1mK above the critical temperature could test theoretical predictions. This goal can be achieved using the existing DECLIC hardware. However, the sample cell will need to be refilled at an average density within 0.1% of the critical density. The proposal research team includes the current CNES DECLIC principal investigator. The commitment from the CNES regarding the refurbishment and operation of the experiment is established. The continuing support from NASA for the utilization of the DECLIC facility is planned to study the critical phenomena, recommended by the National Research Council's (NRC's) decadal survey report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide a unique opportunity to measure experimental properties of a fluid near its liquid-gas critical point without gravitational effects and to refine the theoretical equation of state models. The critical and supercritical state of fluids exhibit unique properties. For instance, the fluid near the critical point and supercritical state shows very large compressibility. The use of these fluids, for instance, for the storage of cryogenic propellant, under normal gravity and/or reduced gravity conditions raises fundamental questions regarding fluid dynamics, heat transfer, interfacial phenomenon. These experimental works will enhance the understanding of basic physics regarding the fluids near critical state.

Task Progress:	During this report period, the new ALI-R insert is successfully tested using the DECLIC facility engineering model at CNES/CADMOS and launched to International Space Station on August 2017. Two science sequences were performed in weightless conditions in ISS to investigate the phase separation phenomena and to confirm the critical parameters. We published two papers on ground-based experiment results. The main experiment to determine the turbidity of the sample is delayed until beginning of 2021 due to the laser source failure in the DECLIC facility. A new DECLIC-EVO facility is currently being planned to replace the DECLIC facility. The ALI-R main science experiment is planned to be performed on early 2021. Among all simple fluids, a striking difference of SF6 data showing a strong deviation from the rectilinear diameter of the coexisting curve is puzzling. To elucidate the problem, we used a simple optical method allowing the high-resolution measurements of the liquid-vapor meniscus position of SF6 as a function of the temperature from far (10 K) to extremely close (1 mK) to the critical temperature. The observed meniscus positions are analyzed using the most recent theoretical approaches to predict consequences of the “complete scaling” and the “law” of the rectilinear diameter. It results that the meniscus position shows a well-defined temperature dependence crossing the volumetric median plane of a highly symmetrical cell, probing unambiguously the absence of the expected deviation in the rectilinear density diameter approaching the SF6 critical point.
Bibliography Type:	Description: (Last Updated: 01/02/2025)
Articles in Peer-reviewed Journals	Garrabos Y, Lecoutre C, Marre S, LeNeindre B, Hahn I. "Liquid-vapor rectilinear diameter revisited." Phys Rev E. 2018 Feb;97(2-1):020101. https://doi.org/10.1103/PhysRevE.97.020101 ; PubMed PMID: 29548230 , Feb-2018
Articles in Peer-reviewed Journals	Lecoutre C, Marre S, Garrabos Y, LeNeindre B, Hahn I. "Near-critical density filling of the SF6 fluid cell for the ALI-R-DECLIC experiment in weightlessness." Acta Astronautica. 2018 May;146:243-8. https://doi.org/10.1016/j.actaastro.2018.03.004 , May-2018